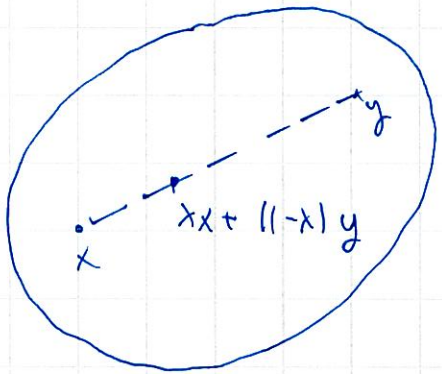
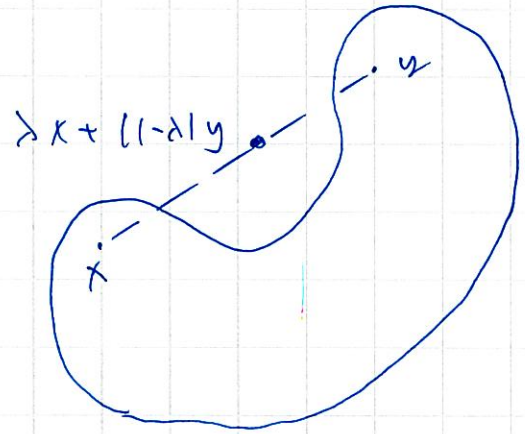
Convex Sets:

Definition: The set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$ and $\lambda \in [0,1]$, we have

$$\lambda x + (1-\lambda) y \in C$$



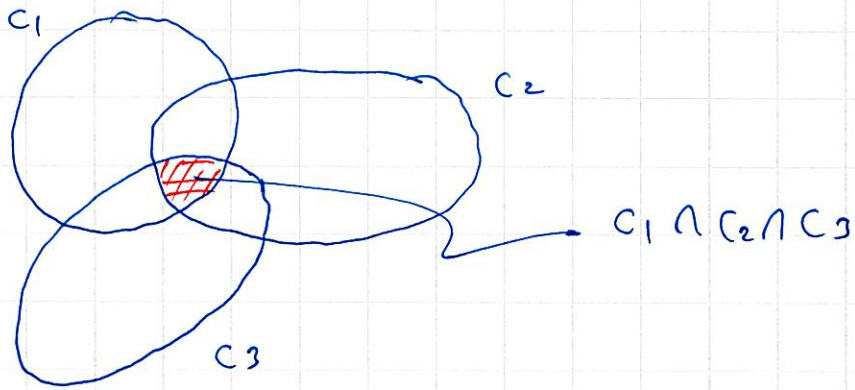
convex



not convex

Intuitively, for a convex set, we have direct line of sight from any point in the set to any other point in the set.

- If $C_1, C_2, \ldots, C_L$ are all convex, then $C_1 \cap C_2 \cap \ldots \cap C_L$ is also convex



Thus, the intersection of bunch of convex sets is still convex

Pf: To show $C_1 \cap C_2 \cap \ldots \cap C_L$ is convex, take any $x, y \in C_1 \cap C_2 \cap \ldots \cap C_L$ and any $\lambda \in [0,1]$, we want to show $\lambda x + (1-\lambda) y \in C_1 \cap C_2 \cap \ldots \cap C_L$.

Because $x \in C_1 \cap C_2 \cap \ldots \cap C_L$, we have $x \in C_1$
$y \in C_1 \cap C_2 \cap \ldots \cap C_L$, we have $y \in C_1$
Because C_1 is convex, $x \in C_1$, and $y \in C_1$, we have $\lambda x + (1-\lambda) y \in C_1$

Similarly, because $x \in C_1 \cap C_2 \cap \ldots \cap C_L$, we have $x \in C_2$
$y \in C_1 \cap C_2 \cap \ldots \cap C_L$, we have $y \in C_2$
Because C_2 is convex, $x \in C_2$, and $y \in C_2$, we have $\lambda x + (1-\lambda) y \in C_2$

Continuing similarly $\lambda x + (1-\lambda) y \in C_3, \ldots \lambda x + (1-\lambda) y \in C_L$
Then we must have $\lambda x + (1-\lambda) y \in C_1 \cap C_2 \cap \ldots \cap C_L$ done!

- Let $f: \mathbb{R}^n \to \mathbb{R}$ be convex, define $G = \{x \in \mathbb{R}^n : f(x) \leq 0\}$

Then G is convex

Pf: Take any $x, y \in G$ and $\lambda \in [0,1]$. We want to show $\lambda x + (1-\lambda) y \in G$

Because $x \in G$, we have $f(x) \leq 0$
$y \in G$, we have $f(y) \leq 0$

Thus,

$$\begin{array}{rl} f(x) \leq 0 & * \lambda \\ + \; f(y) \leq 0 & * (1-\lambda) \\ \hline \lambda f(x) + (1-\lambda) f(y) \leq 0 & \end{array}$$

Also, because f is convex $f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$

So we get $f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y) \leq 0$

Because $f(\lambda x + (1-\lambda) y) \leq 0$, we have $\lambda x + (1-\lambda) y \in G$ done!

- Why are these two results useful? Consider the constrained optimization problem:

$$\min \quad f(x)$$
$$\text{st} \quad g_1(x) \le 0, \quad g_2(x) \le 0, \ldots, \quad g_m(x) \le 0,$$

where $g_1, g_2, \ldots, g_m$ are convex functions

- The set of feasible solutions for this problem is

$$C = \{ x \in \mathbb{R}^n : \ g_1(x) \le 0, \ g_2(x) \le 0, \ldots, \ g_m(x) \le 0 \}$$

- Define $\quad C_i = \{ x \in \mathbb{R}^n : \ g_i(x) \le 0 \}$

Because g_i is convex, C_i is convex for all $i = 1, \ldots, m$

- Also $\quad C = C_1 \cap C_2 \cap \ldots \cap C_m$

Because $C_1, \ldots, C_m$ are all convex, C is convex as well because C is the intersection of a bunch of convex sets

- Thus, if we have an optimization problem with left side of all "$\le$" constraints are convex functions, then the set of feasible solutions is convex set

Projected gradient Descent

Definition: The function $f: \mathbb{R}^n \to \mathbb{R}$ is strictly convex if
$$f(\lambda x + (1-\lambda) y) < \lambda f(x) + (1-\lambda) f(y) \quad \text{for all } x, y \in \mathbb{R}^n \ x \neq y, \ \lambda \in (0,1)$$

- Let $g: \mathbb{R}^n \to \mathbb{R}$ be a strictly convex function and X be a convex set. Consider the constrained optimization problem

$$\min_{x \in X} g(x)$$

Then, there is a unique optimal solution to this problem.

(Proof on next page)

Pf: To get a contradiction, let x^* and y^* with $x^* \neq y^*$ be two optimal solutions to the problem above. Let Z^* be the opt. obj. value

So, $Z^* = \min_{x \in X} g(x)$

- Because x^* is an opt. solution, we have $x^* \in X$ and $g(x^*) = Z^*$
  Because y^* is an opt. solution, we have $y^* \in X$ and $g(y^*) = Z^*$

- Because $x^* \in X$, $y^* \in X$ and X is convex, we have $\lambda x^* + (1-\lambda) y^* \in X$.

- Because g is strictly convex, $g(x^*) = Z^*$ and $g(y^*) = Z^*$, we have

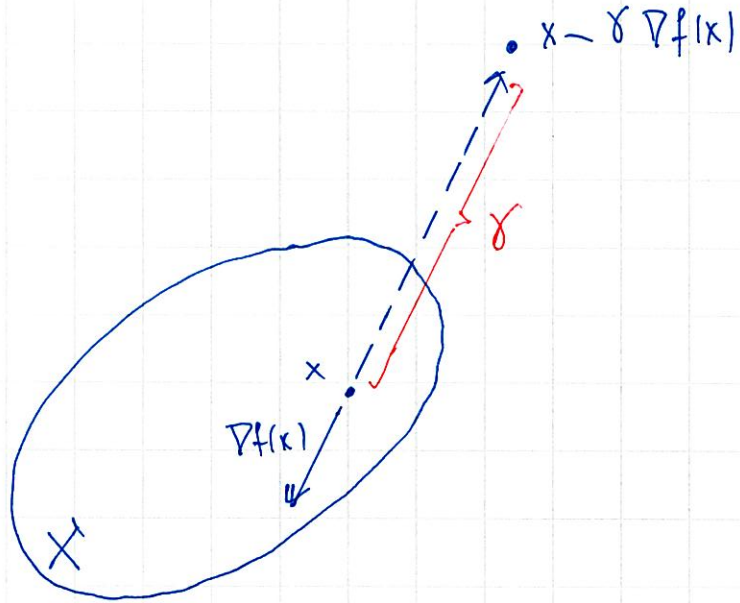
$$g(\lambda x^* + (1-\lambda) y^*) < \lambda \underbrace{g(x^*)}_{Z^*} + (1-\lambda) \underbrace{g(y^*)}_{Z^*} = Z^*$$

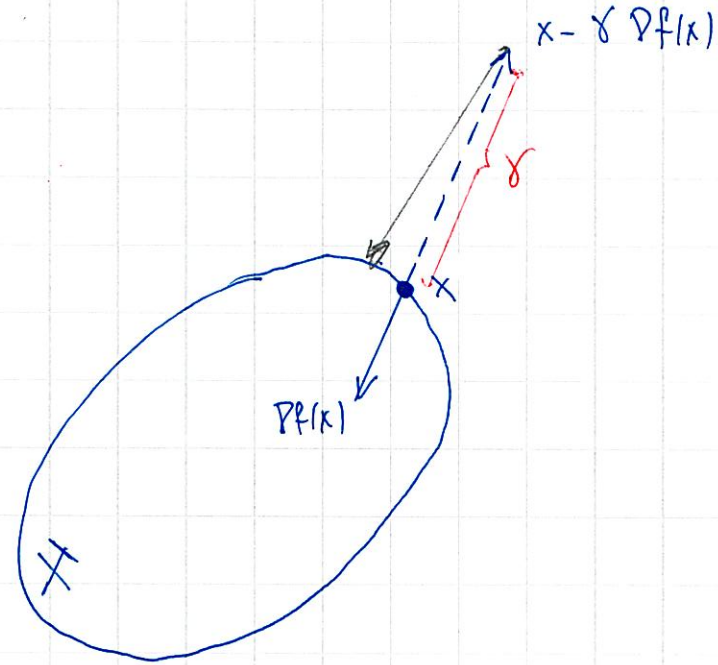
- Thus, $\lambda x^* + (1-\lambda) y^* \in X$ so $\lambda x^* + (1-\lambda) y^*$ is feasible to the opt. pro
  Also $g(\lambda x^* + (1-\lambda) y^*) < Z^*$, so $\lambda x^* + (1-\lambda) y^*$ is giving an obj. value strictly better than Z^*, which is the obj. value from x^* or y^*.

So x^* and y^* cannot be an optimal solution. Contradiction

Done!

- When we apply gradient descent for a constrained opt. problem, moving against the gradient can take us outside the set of feasible solutions.



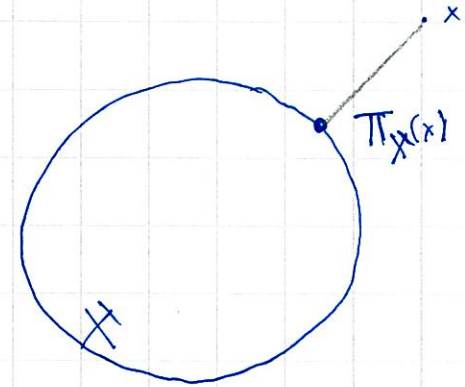


- Projected gradient descent is useful for solving constrained optimization problem. Assume that we are currently at point x and move against the gradient for a step size of γ to reach the point $x - \gamma \nabla f(x)$.

- If $x - \gamma \nabla f(x)$ is not in the feasible region X, then we project $x - \gamma \nabla f(x)$ back on to X.
  That is, we find the point in X that is closest to $x - \gamma f(x)$

- Define the projection of x onto the convex set $\mathcal{X}$ as

$$\Pi_{\mathcal{X}}(x) = \underset{y \in \mathcal{X}}{\arg\min} \; \|y - x\|^2$$



- Note if $x \in \mathcal{X}$, then $\Pi_{\mathcal{X}}(x) = x$.

- The projected gradient descent works as follows. We start with some initial point x^0. At any iteration k, we do the following

$$y^{k+1} = x^k - \gamma \nabla f(x^k)$$

$$x^{k+1} = \Pi_{\mathcal{X}}(y^{k+1})$$